National University of Computer and Emerging Sciences

Data Mining (DS3002)

Date: April 3rd 2024

Course Instructor

Miss Eesha Tur Razia Babar

Sessional-I Exam

Total Time: 1 Hours Total Marks: 25

Total Questions: 05

Semester: Spring-2024

Campus: Lahore

Dept: AI and Data Science

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Student Name Roll No Section Student Signature

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CLO 1: Understand basic concepts of data mining.

Q1: Suppose that we have training data f(x(1); y(1)); (x(2); y(2)); : : : ; (x(m); y(m)) and we wish to predict y using a nonlinear regression model with two parameters:
y (pred) = a + exp(x1 + b)

We decided to train our model using gradient descent on the mean squared error (MSE).

1a) Write down the expression for the MSE on our training set. (1 mark)
= 1/m summation (y – a + exp(x1 + b)^2
1b) Write down the gradient of the MSE. (3 mark)

Wrp to a =
=-2/m (y – a + exp(x1 + b))
Wrp to be
= -2/m (y – a+exp(x1+b))e(x1+b)

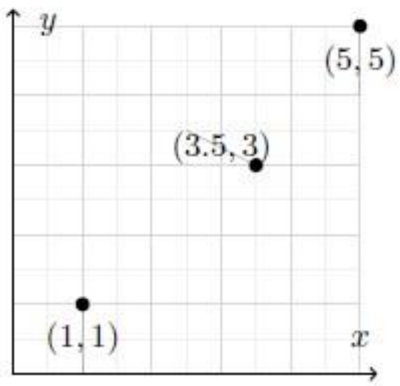
1c) Find leave one out cross-validation error of a linear predictor using the following points. (3 marks)

	Pred, $\hat{y}$	true, y
a)	$-\frac{1}{3}$	1
b)	3.5	3
c)	4.2	5

$$MSE = \frac{1}{3}\left(\left(-\frac{1}{3}-1\right)^2 + (3.5-3)^2 + (4.2-5)^2\right)$$

$$= \frac{1}{3}\left(\frac{16}{9} + \frac{1}{4} + \frac{16}{25}\right)$$

$$\sim 0.889$$



CLO #2: Apply data mining techniques to extract information from large data sets

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Q2: FP Growth ***[10 marks]***

Generate frequent patterns through FP growth using the following transaction data set and min_support = 2. Please show complete work. You don't have to draw the separate conditional FP trees.

Transaction ID	List of items in the transaction
T1	B , A , T
T2	A , C
T3	A , S
T4	B , A , C
T5	B , S
T6	A , S
T7	B , S
T8	B , A , S , T
T9	B , A , S

2 marks

Item	Support Count
Beans (B)	6
Asparagus (A)	7
Squash (S)	6
Corn (C)	2
Tomatoes (T)	2

Item	Support Count
Asparagus (A)	7
Beans (B)	6
Squash (S)	6
Corn (C)	2
Tomatoes (T)	2

Rearrange Table 2 marks
Construct FP Tree 2 marks
Conditional FP Tree 2 marks
Frequent Pattern Generation 2 marks

Item	Conditional Pattern base	Conditional FP tree	Frequent Pattern Generation
Tomatoes (T)	{{A,B:1},{A,B,S:1}}	<A:2,B:2>	{A,T:2},{B,T:2},{A,B,T:2}
Corn (C)	{{A,B:1},{A:1}}	<A:2>	{A,C:2}
Squash (S)	{{A,B:2},{A:2},{B:2}}	<A:4,B:2>,<B:2>	{A,S:4},{B,S:4},{A,B,S:2}
Bean (B)	{{A:4}}	<A:4>	{A,B:4}

CLO #2: Apply data mining techniques to extract information from large data sets

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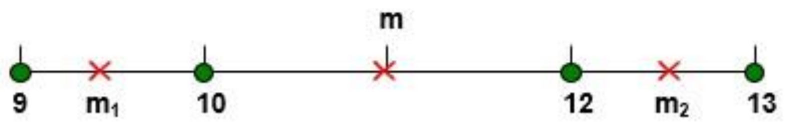
Q3: Regular sunscreen use can reduce the risk of skin cancer. A 2015 study asked a large random sample of American adults about their use of sunscreen on their face and on other exposed skin when going out in the sun. The study found that 18.1% of the men surveyed regularly use sunscreen

on the face, 19.9% regularly use sunscreen on other exposed skin, and 14.3% regularly use sunscreen on both the face and other exposed skin.
Probability model for sunscreen use in the population of American adult men going out in the sun: SF "regularly use sunscreen on the face" SO "regularly use sunscreen on other exposed skin"

1) *P*(SF or SO) (1 mark)
   = P(SF) + P(SO) – P(SF AND SO)
   = 0.181+0.199-0.143
   = 0.237
2) Are SF and SO independent? (2 mark)
   = P(SF and S0) = P (SF) P (SO)
   0.143 is not equal to 0.036
   SF and SO are not independent.

CLO #2: Apply data mining techniques to extract information from large data sets

Q4: 2 marks



Find SSE and SSB for K = 2 clusters using the above data.
SSE = 1
SSB = 9

CLO #2: Apply data mining techniques to extract information from the large data sets

Q5: True/False [1+1+1 = 3 marks]
Please write all answers on answer sheet.

1) **An item is maximal frequent if none of its immediate supersets is frequent. True**
2) **To find cross-support patterns we compute the relationship between min support and min confidence. False**
3) **According to the Apriori algorithm if an item set is frequent then all its supersets will also be frequent. False**